

Meshtastic SOS Keyboard

Overview

The **Meshtastic SOS Keyboard** is a standalone emergency messaging device designed for reliable offline communication over the Meshtastic mesh network.

It allows the user to:

- Send predefined emergency messages
- Use physical buttons for fast actions
- Operate without internet
- Run continuously as a background service
- Automatically reconnect to Meshtastic nodes
- Confirm message delivery using ACK feedback
- Support multiple Meshtastic devices automatically

The project is designed to be:

- Simple
- Reliable
- Low-maintenance
- Community-friendly
- Easy to reproduce

Main Features

Emergency Button System

Each physical button can send a predefined message.

Example:

Button		Message
GPIO17		Emergência
GPIO27		Podem ligar?
GPIO22		Está tudo bem
GPIO23		Quando puderes, passa cá

Messages are fully customizable in `config.py`.

Long Press Protection

Buttons only trigger after a configurable long press.

Default:

```
LONG_PRESS_SECONDS = 2
```

This prevents accidental transmissions.

Automatic Meshtastic Reconnection

If the USB Meshtastic node disconnects:

- the software automatically retries connection
- no reboot required
- works after power failures or cable reconnects

ACK Confirmation System

The device waits for Meshtastic ACK confirmations.

Success

If ACK is received:

- green LED
- short buzzer beep

Failure

If ACK is not received:

- red LED
- long buzzer beep

Multi-Device Support

The software automatically searches for available Meshtastic devices.

Supported examples:

- LilyGO T-Beam
- T-LoRa
- Heltec V3
- RAK WisBlock
- USB serial Meshtastic nodes

No fixed serial port required.

Hardware Requirements

Raspberry Pi

Compatible with:

- Raspberry Pi Zero 2 W
- Raspberry Pi 3
- Raspberry Pi 4
- Raspberry Pi 5

Required Components

Component	Required
Raspberry Pi	Yes
Meshtastic node (USB)	Yes
Push buttons	Yes
RGB LED	Optional
Buzzer	Optional
Battery UPS	Optional

GPIO Wiring

Buttons

Each button connects:

GPIO ----- BUTTON ----- GND

Internal pull-ups are enabled automatically.

No external resistor required.

RGB LED

Example wiring:

LED Pin Raspberry GPIO

Red GPIO5

Green GPIO6

Blue GPIO13

Common GND or 3.3V

Important

Use resistors:

220Ω to 330Ω

One resistor per color channel.

Buzzer

Buzzer Pin Raspberry GPIO

Signal GPIO18

GND GND

LED Status Codes

Color	Meaning
Blue	Booting
Purple	Connecting to Meshtastic
Green	Connected / ACK success
Yellow	Sending message
Red	Error / ACK failure

Buzzer Codes

Sound	Meaning
Short beep	Success
Long beep	Error
Double short beep	Startup

Troubleshooting

No Meshtastic Device Found

Check USB

`meshtastic --info`

If not detected:

- replace USB cable
- verify Meshtastic firmware
- reconnect device

Buttons Not Working

Check GPIO

Verify wiring:

`GPIO -> BUTTON -> GND`

RGB LED Always On

Possible causes:

- common anode vs common cathode mismatch
- wrong polarity
- missing resistor

Safety Notes

This project is intended as:

- emergency aid tool
- offline communication system
- community resilience platform

It should not replace:

- certified emergency systems
- official rescue equipment
- medical alert systems

Always test your mesh network coverage before real-world deployment.

License

Recommended license:

MIT License

Simple, open-source, community-friendly.

Credits

Built using:

- [Meshtastic Portugal](#)
- [Meshtastic Official Website](#)
- [Meshtastic Python API](#)
- [gpiozero Documentation](#)
- [Raspberry Pi Official Website](#)